

Project Overview

Topic: Neurosymbolic Reasoning

Huỳnh Gia Bảo, Nguyễn Ngọc Cảnh, Nguyễn Đức Trí,
Phan Huỳnh Minh Khôi, Trần Lý Nhật Hào

Faculty of Information Technology
University of Science - VNUHCM

January 28, 2026

Contents

- 1 Motivation for Neurosymbolic AI
- 2 Problem Statement
- 3 Data
- 4 Related Work
- 5 Challenges & Solutions

Contents

- 1 Motivation for Neurosymbolic AI
- 2 Problem Statement
- 3 Data
- 4 Related Work
- 5 Challenges & Solutions

Classical AI: Rule-based/Symbolic AI

Knowledge Base

$\forall x : \text{Human}(x) \rightarrow \text{Wibu}(x)$
(All humans are wibu.)

$\text{Human}(\text{Huy})$
(Huy is human.)

Classical AI: Rule-based/Symbolic AI

Knowledge Base

$\forall x : \text{Human}(x) \rightarrow \text{Wibu}(x)$
(All humans are wibu.)

$\text{Human}(\text{Huy})$
(Huy is human.)

Deduction Mechanism

$A \rightarrow B$	
A	
$\therefore B$	(Syllogism/Modus ponens)

Classical AI: Rule-based/Symbolic AI

Knowledge Base

$\forall x : \text{Human}(x) \rightarrow \text{Wibu}(x)$
(All humans are wibu.)

$\text{Human}(\text{Huy})$
(Huy is human.)

Deduction Mechanism

$A \rightarrow B$	
A	
$\therefore B$	(Syllogism/Modus ponens)

Conclusion

$\text{Wibu}(\text{Huy})$
(Huy is wibu.)

Classical AI: Rule-based/Symbolic AI

Knowledge Base

$\forall x : \text{Human}(x) \rightarrow \text{Wibu}(x)$
(All humans are wibu.)

$\text{Human}(\text{Huy})$
(Huy is human.)

Deduction Mechanism

$A \rightarrow B$
 A
—
 $\therefore B$

(Syllogism/Modus ponens)

Conclusion

$\text{Wibu}(\text{Huy})$
(Huy is wibu.)

Pros

- Explainability
- Data Efficiency
- Abstract Reasoning

Classical AI: Rule-based/Symbolic AI

Knowledge Base

$\forall x : \text{Human}(x) \rightarrow \text{Wibu}(x)$
(All humans are wibu.)

$\text{Human}(\text{Huy})$
(Huy is human.)

Deduction Mechanism

$A \rightarrow B$
 A
—
 $\therefore B$

(Syllogism/Modus ponens)

Conclusion

$\text{Wibu}(\text{Huy})$
(Huy is wibu.)

Pros

- Explainability
- Data Efficiency
- Abstract Reasoning

Cons

- Brittleness
- Scalability
- Poor Perception

Classical AI: Rule-based/Symbolic AI

Knowledge Base

$\forall x : \text{Human}(x) \rightarrow \text{Wibu}(x)$
(All humans are wibu.)

$\text{Human}(\text{Huy})$
(Huy is human.)

Deduction Mechanism

$A \rightarrow B$
 A

 $\therefore B$

(Syllogism/Modus ponens)

Conclusion

$\text{Wibu}(\text{Huy})$
(Huy is wibu.)

Pros

- Explainability
- Data Efficiency
- Abstract Reasoning

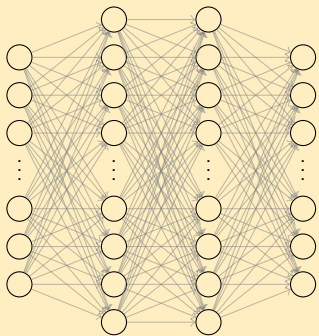
Cons

- Brittleness
- Scalability
- Poor Perception

⇒ **reliable but inflexible.**

Modern AI: Data-driven/Neural Networks

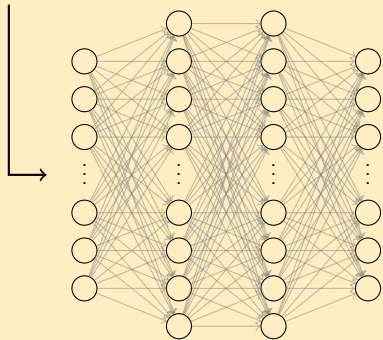
Neural Networks



Modern AI: Data-driven/Neural Networks

Neural Networks

“Is Huy a good boy?”

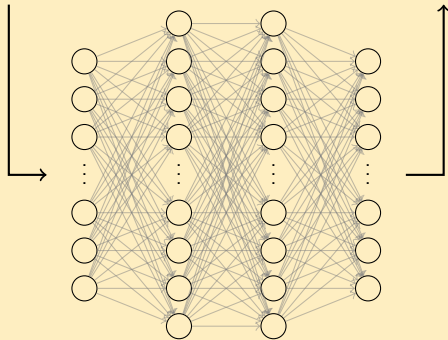


Modern AI: Data-driven/Neural Networks

Neural Networks

“Is Huy a good boy?”

“No, he’s wibu.”

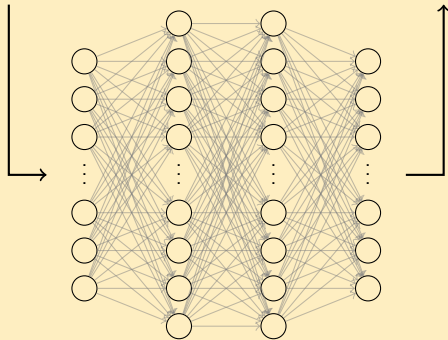


Modern AI: Data-driven/Neural Networks

Neural Networks

"Is Huy a good boy?"

"No, he's wibu."



Pros

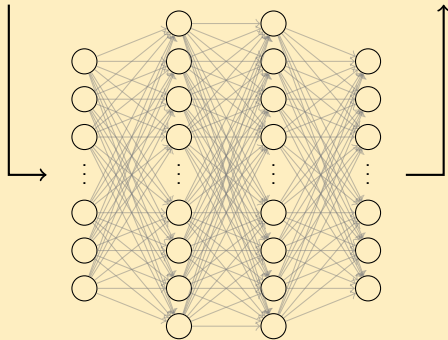
- Generalization & Robustness
- Perception
- Automated Feature Extraction

Modern AI: Data-driven/Neural Networks

Neural Networks

"Is Huy a good boy?"

"No, he's wibu."



Pros

- Generalization & Robustness
- Perception
- Automated Feature Extraction

Cons

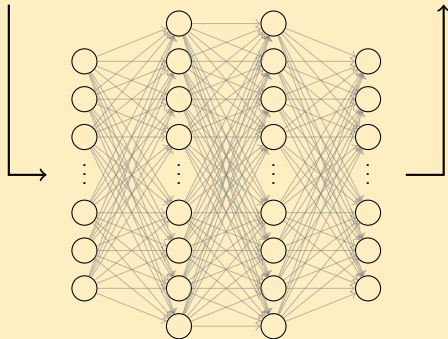
- Lack of Explainability
- Data & Resource Hungry
- Hallucination

Modern AI: Data-driven/Neural Networks

Neural Networks

"Is Huy a good boy?"

"No, he's wibu."



Pros

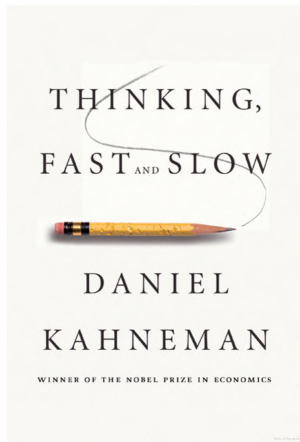
- Generalization & Robustness
- Perception
- Automated Feature Extraction

Cons

- Lack of Explainability
- Data & Resource Hungry
- Hallucination

⇒ **flexible but unreliable.**

Neurosymbolic AI: A Hybrid Approach



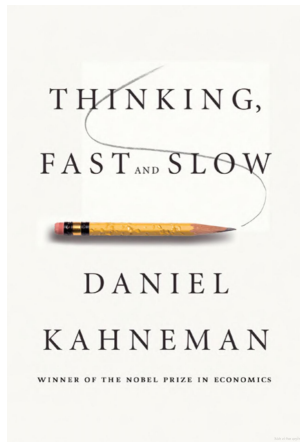
SYSTEM 1 (FAST)

Intuition & Instinct

SYSTEM 2 (SLOW)

Rational thinking

Neurosymbolic AI: A Hybrid Approach



SYSTEM 1 (FAST)

Intuition & Instinct

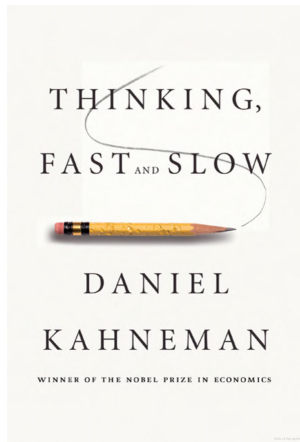
SYSTEM 2 (SLOW)

Rational thinking

||

NEURAL NETWORK

Neurosymbolic AI: A Hybrid Approach



SYSTEM 1 (FAST)

Intuition & Instinct

||

NEURAL NETWORK

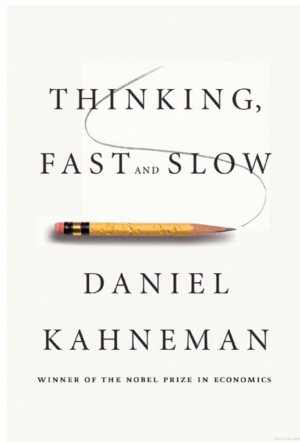
SYSTEM 2 (SLOW)

Rational thinking

||

SYMBOLIC ENGINE

Neurosymbolic AI: A Hybrid Approach



SYSTEM 1 (FAST)

Intuition & Instinct

||

NEURAL NETWORK

+

SYSTEM 2 (SLOW)

Rational thinking

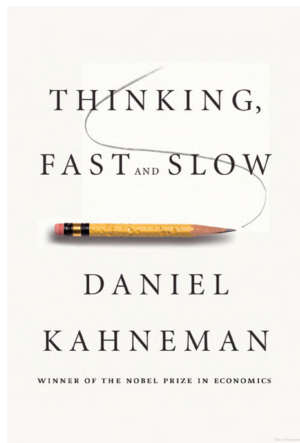
||

SYMBOLIC ENGINE

=

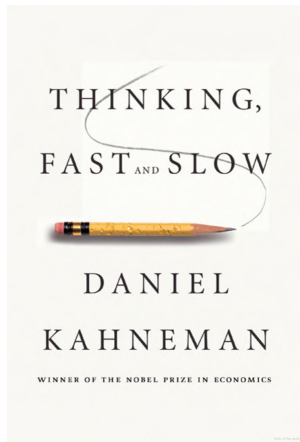
**HUMAN
INTELLIGENCE**

Neurosymbolic AI: A Hybrid Approach



NEURAL NETWORK $\oplus$ SYMBOLIC ENGINE = NEUROSYMBOLIC AI

Neurosymbolic AI: A Hybrid Approach



Semantic Parsing/Language Formalization

NEURAL NETWORK $\oplus$ SYMBOLIC ENGINE = NEUROSYMBOLIC AI

Contents

- 1 Motivation for Neurosymbolic AI
- 2 Problem Statement**
- 3 Data
- 4 Related Work
- 5 Challenges & Solutions

Problem Statement

Language Formalization

The goal is to learn a mapping function $f: \mathcal{X} \rightarrow \mathcal{Y}$ that translates Natural Language sentences (x_{NL}) into machine-verifiable First-Order Logic formulas (y_{FOL}).

Problem Statement

Language Formalization

The goal is to learn a mapping function $f: \mathcal{X} \rightarrow \mathcal{Y}$ that translates Natural Language sentences (x_{NL}) into machine-verifiable First-Order Logic formulas (y_{FOL}).

Input (x_{NL})

*Natural Language
(Unstructured)*

"Every student takes a course."

$f(\cdot)$
→

Output (y_{FOL})

*First-Order Logic
(Structured)*

$\forall x(\text{Student}(x) \rightarrow \exists y(\text{Course}(y) \wedge \text{Takes}(x, y)))$

Problem Statement

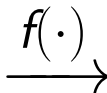
Language Formalization

The goal is to learn a mapping function $f: \mathcal{X} \rightarrow \mathcal{Y}$ that translates Natural Language sentences (x_{NL}) into machine-verifiable First-Order Logic formulas (y_{FOL}).

Input (x_{NL})

*Natural Language
(Unstructured)*

"Every student takes a course."



Output (y_{FOL})

*First-Order Logic
(Structured)*

$\forall x(\text{Student}(x) \rightarrow \exists y(\text{Course}(y) \wedge \text{Takes}(x, y)))$

Core Challenges:

- **The Requirement for Absolute Precision:** Formal languages are syntactically fragile. A formal expression must be 100% correct to be machine-verifiable.
- **Data Scarcity and the Expert-Data Bottleneck:** Creating NL-FOL pairs requires professionals with deep knowledge of both the domain and the formal syntax.

Contents

- 1 Motivation for Neurosymbolic AI
- 2 Problem Statement
- 3 Data**
- 4 Related Work
- 5 Challenges & Solutions

Data Source

The data used consists of the **training sets** of the datasets listed in the table below.

Source	Volume	Description
FOLIO	1,000 samples	With FOL label , with true/false/uncertain label.
ProofWriter	41,400 samples	Without FOL label , with true/false label, with proof.
RuleTaker	480,000 samples	Without FOL label , with entailment/not entailment label.
ReClor	4,637 samples	Without FOL label , in multiple-choice format.
LogiQA	7,380 samples	Without FOL label , in multiple choice format.

Table: Preliminary statistics of data sources.

Data Source

FOLIO Dataset

FOLIO Dataset

FOLIO is an expert-written, open-domain, logically complex and diverse dataset for natural language reasoning with first-order logic.

Examples:

Premises	Premises-FOL	Conclusion	Conclusion-FOL	Label
All rabbits have fur. Some pets are rabbits.	$\forall x (\text{Rabbit}(x) \rightarrow \text{Have}(x, \text{fur}))$ $\exists x (\text{Pet}(x) \wedge \text{Rabbit}(x))$	Some pets do not have fur.	$\exists x \exists y (\text{Pet}(x) \wedge \text{Pet}(y) \wedge \neg \text{Have}(x, \text{fur}) \wedge \neg \text{Have}(y, \text{fur}))$	Uncertain
All tigers are cats. No cats are dogs. All Bengal tigers are tigers. All huskies are dogs. Fido is either a Bengal tiger or a cat.	$\forall x (\text{Tiger}(x) \rightarrow \text{Cat}(x))$ $\forall x (\text{Cat}(x) \rightarrow \neg \text{Dog}(x))$ $\forall x (\text{BengalTiger}(x) \rightarrow \text{Tiger}(x))$ $\forall x (\text{Husky}(x) \rightarrow \text{Dog}(x))$ $\text{BengalTiger}(\text{fido}) \oplus \text{Cat}(\text{fido})$	Fido is neither a dog nor a husky.	$\neg \text{Dog}(\text{fido}) \wedge \neg \text{Husky}(\text{fido})$	True
...	...	...	...	...

Table: Sample data from the FOLIO dataset.

Data Source

ProofWriter Dataset

ProofWriter Dataset

ProofWriter is ...

Examples:

Data Source

RuleTaker Dataset

RuleTaker Dataset

RuleTaker is ...

Examples:

Data Source

ReClor Dataset

ReClor Dataset

ReClor is a reading comprehension dataset for assessing logical reasoning in LLMs via context-based multiple-choice tasks.

Examples:

Context	Question	Answers	Label
Shipping Clerk: The five specially ordered shipments sent out last week were sent out on Thursday. Last week, all of the shipments that were sent out on Friday consisted entirely of building supplies, and the shipping department then closed for the weekend. Four shipments were sent to Truax Construction last week, only three of which consisted of building supplies.	If the shipping clerk's statements are true, which of the following must also be true?	0. At least one of the shipments sent to Truax Construction last week was sent out before Friday. 1. At least one of last week's specially ordered shipments did not consist of building supplies. 2. At least one of the shipments sent to Truax Construction last week was specially ordered. 3. At least one of the shipments sent to Truax Construction was not sent out on Thursday of last week.	0
...	...	...	...

Table: Sample data from the ReClor dataset.

Data Source

LogiQA Dataset

LogiQA Dataset

LogiQA is ...

Examples:

Data Pipeline

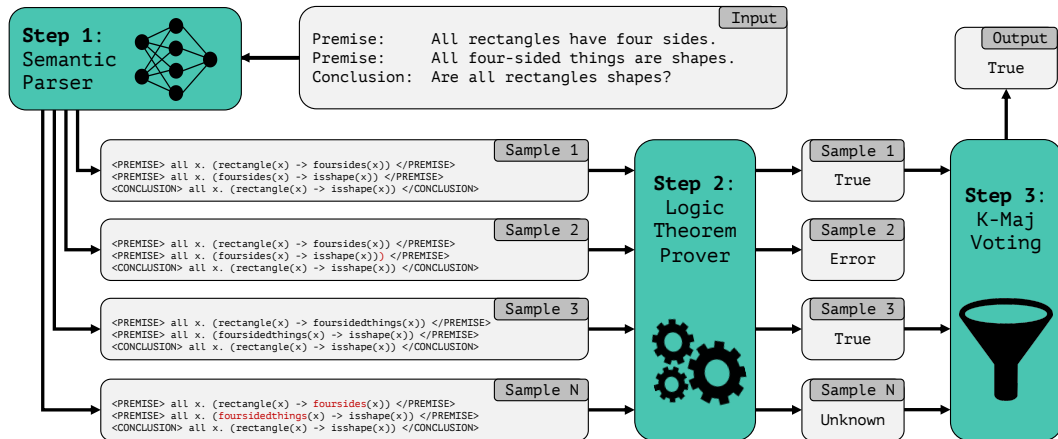


Figure: Mình không hẳn là dùng pipeline này, để tạm đây mai sửa

Data Pipeline

(Viết chi tiết aabcxyz về data pipeline, giai đoạn 1 dùng SFT, giai đoạn 2 dùng DPO)

Supervised Fine-Tuning (SFT):

$$L_{\text{SFT}}(\theta) = - \sum \log \mathbb{P}(y_{\text{FOL}} \mid x_{\text{NL}}).$$

Direct Preference Optimization (DPO):

$$L_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l)} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi_{\theta}(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right) \right].$$

Contents

- 1 Motivation for Neurosymbolic AI
- 2 Problem Statement
- 3 Data
- 4 Related Work**
- 5 Challenges & Solutions

Related Work

Liệt kê ngắn gọn đóng góp của các paper liên quan

Contents

- 1 Motivation for Neurosymbolic AI
- 2 Problem Statement
- 3 Data
- 4 Related Work
- 5 Challenges & Solutions**

Anticipated Challenges & Proposed Solutions

Challenge X

XXX

Solution Y

YYY

Thank you!